

❑ Comparative Evaluation of Grok-3 Model in Multi-Language Battles

- Dataset: Multi-model battle evaluations
- Focus: Measuring Grok-3's strengths and weaknesses across languages and other models
- Tools used: Python, Pandas, Matplotlib, Seaborn
- Prepared by: *Grok Team*

This project aims to understand how Grok performs compared to other LLMs in real evaluation battles.

❑ Dataset Structure and Features

- 24 columns representing model interactions and outcomes and 5180 record
- Each row = one **battle** between two models
- Key columns: model_a, model_b, winner, language, user_prompt, response_a, response_b
- Additional flags: grok_in_a, grok_in_b, timestamp

The dataset captures both the content and outcome of model-to-model evaluations.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5180 entries, 0 to 5179
Data columns (total 24 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   id                                    5180 non-null   object 
 1   model_a                              5180 non-null   object 
 2   model_b                              5180 non-null   object 
 3   winner                               5180 non-null   object 
 4   evaluation_session_id                5180 non-null   object 
 5   evaluation_order                     5180 non-null   int64  
 6   conversation_a                       5180 non-null   object 
 7   conversation_b                       5180 non-null   object 
 8   full_conversation                    5180 non-null   object 
 9   conv_metadata                        5180 non-null   object 
10   category_tag                         5180 non-null   object 
11   language                             5180 non-null   object 
12   is_code                              5180 non-null   bool    
13   timestamp                            5180 non-null   object 
14   turns                                5180 non-null   int64  
15   response_a                           5179 non-null   object 
16   response_b                           5178 non-null   object 
17   user_prompt                          5180 non-null   object 
18   gpt_in_a                             5180 non-null   bool    
19   gpt_in_b                             5180 non-null   bool    
20   gemini_in_a                          5180 non-null   bool    
21   gemini_in_b                          5180 non-null   bool    
22   grok_in_a                            5180 non-null   bool    
23   grok_in_b                            5180 non-null   bool    
dtypes: bool(7), int64(2), object(15)
memory usage: 723.5+ KB
```

❑ How the Battles Were Evaluated

- Each battle compares Grok-3 against another model (Gemini, Claude, GPT, etc.)
- “winner” column defines which response was preferred

Analytical objectives:

1. Compute Grok’s win rate per opponent
2. Compute Grok’s win rate per language
3. Qualitatively inspect loss cases

- Grok’s results were analyzed both quantitatively and qualitatively.

❑ Classification of Battle Results (Winner column)

- **model_a** → Model A wins
- **model_b** → Model B wins
- **tie** → Equal performance
- **both_bad** → Both responses unsatisfactory

Only the first three were used for the main analysis

“Both bad” results (~10%) were excluded from win-rate analysis to avoid bias.

❑ Global Distribution of Grok Battles

- Total battles analyzed: ≈ 5180
- Wins: 2476
- Losses: 1875
- Ties: 829
- Both bad: 517 (excluded)
- Overall win rate $\approx 48\%$

Conclusion:

Grok performs competitively but not dominantly across all opponents.

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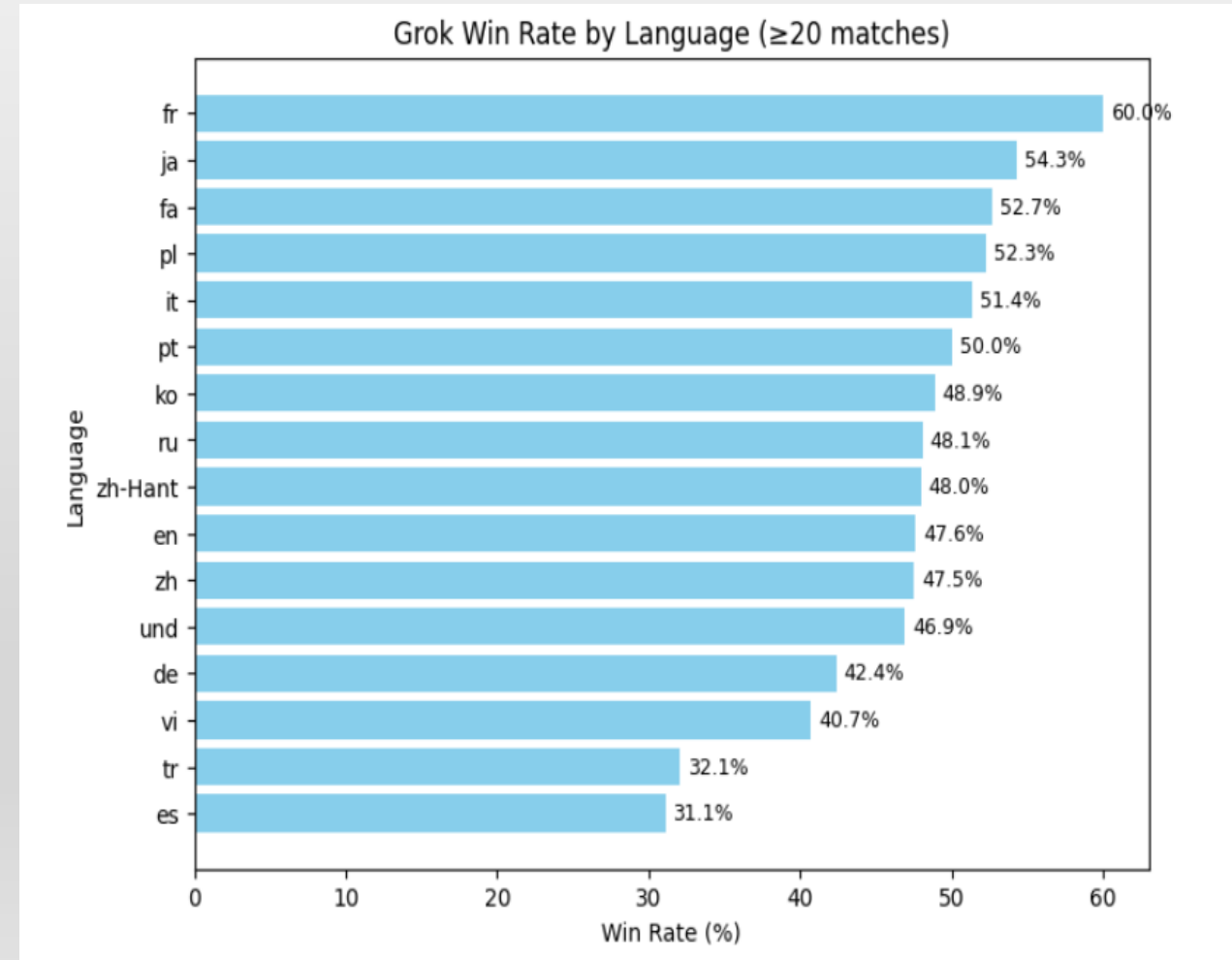
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❑ Grok Performance Across Languages

- Highest win rates: French, Polish, Japanese
- Moderate: Persian ($\approx 52\%$), Korean ($\approx 49\%$)
- Weakest: Spanish, Turkish, Arabic
- Only languages with ≥ 30 matches were considered

Conclusion:

Grok's multilingual ability is strong overall but uneven across regions.



❑ Models Where Grok Struggled

- Focused subset: Opponents with Grok win rate < 35 %
- Typical rivals: Claude-3-Sonnet, Gemini 2.5 Pro, O3-series
- Analysis limited to battles Grok lost against these models
- **Next step:** Inspect prompts and responses qualitatively

Conclusion:

These models consistently outperform Grok in reasoning and clarity tasks.

	opponent	matches	wins	ties	losses	win_rate(%)
2	chatgpt-4o-latest-20250326	155	54	30	54	34.8
12	deepseek-r1-0528	123	42	17	50	34.1
19	gemini-2.5-pro	171	42	34	75	24.6
20	gemini-2.5-pro-preview-03-25	44	13	10	17	29.5
21	gemini-2.5-pro-preview-05-06	49	11	8	25	22.4
41	o3-2025-04-16	158	50	31	54	31.6

❑ Quantitative Analysis — Persian Losses

Metric	Grok	Opponent	Observation
Avg. response length	2916	1845	Grok answers are longer
Avg. line count	39.9	33.2	Grok is more verbose
List-style responses (%)	72.7 %	63.6 %	Both structured; Grok overuses lists
Avg. prompt length (chars)	123	—	Short prompts

❑ Quantitative Analysis — English Losses

Metric	Grok	Opponent	Observation
Avg. Response Length	5149 chars	3273 chars	Grok's replies are about 57 % longer → tendency toward verbosity
Avg. Line Count	79.5 lines	62.1 lines	Grok writes more paragraphs → less concise structure
List-Style Responses (%)	84.8 %	78.0 %	Both use lists, but Grok relies on enumeration more heavily
Avg. Prompt Length (chars)	540.9	—	English prompts are roughly 4× longer than Persian ones
Avg. Prompt Length (words)	69.1	—	Complex, multi-step instructions are common

❑ Quantitative Analysis — English Losses

In English battles, Grok produces significantly longer and more structured answers than its opponents,

yet it still loses in overall evaluation.

The results indicate that Grok's **verbosity does not translate to higher quality** — responses often include redundant reasoning or loosely connected facts.

Opponent models (Claude, Gemini, O-series) typically provide **shorter, logically cohesive** explanations that better follow the prompt's intent.

The longer prompts suggest higher reasoning complexity, exposing Grok's difficulty in **maintaining coherence over extended instructions**.

Conclusion:

Grok's English responses are fluent and rich in detail but lose due to redundancy and limited reasoning precision.

❑ Persian vs English — Comparative Summary

Feature	Persian	English	Interpretation
Response length	Longer	Shorter	Persian answers verbose
Structure (lists %)	73 %	~55 %	Persian more structured
Prompt length	Short	Long	English prompts more detailed
Reasoning accuracy	Lower	Higher	Better reasoning in English

- Grok writes fluently in Persian but shows weaker reasoning control compared to English.

❑ Summary of Insights

- Grok's global win rate $\approx 48\%$
- Performs well in structured factual queries
- Weak in concise reasoning tasks
- Persian outputs are verbose and less precise
- English outputs are concise but still lose in reasoning depth

Conclusion:

Grok balances fluency and verbosity but needs reasoning-focused tuning.